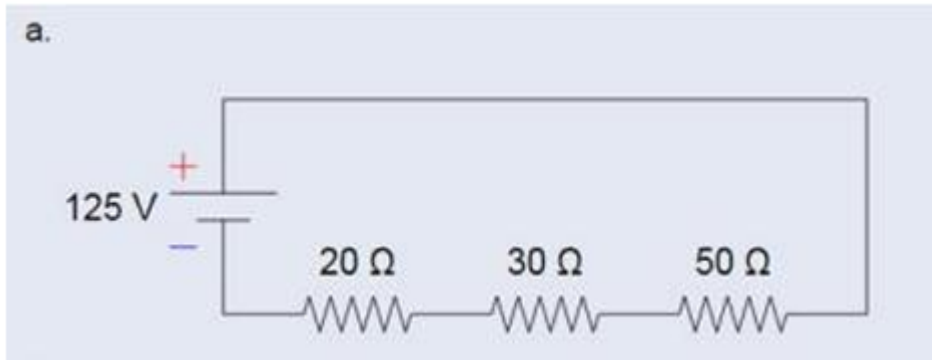


168263 ~ 1CS 4E



1. Total Resistance in Ohms:

Given:  $V = 125 \text{ V}$        $R_3 = 50 \Omega$

$R_1 = 20 \Omega$

$R_2 = 30 \Omega$

Total resistance in series,  $R_T = R_1 + R_2 + R_3$

$$R_T = 20 + 30 + 50$$

$$= 100 \Omega$$

2. Total current

$$V = RI$$

$$125 = 100I$$

$$I = 1.25 \text{ A}$$

3. Current through each resistor

$$I_1 = I_2 = I_3 = I_T \\ = 1.25 \text{ A}$$

4. Voltage Drop per Resistor

$$V = RI$$

$$V_1 = 20 \times 1.25 \\ = 25 \text{ V}$$

$$V_3 = 50 \times 1.25 \\ = 62.5 \text{ V}$$

$$V_2 = 30 \times 1.25 \\ = 37.5 \text{ V}$$

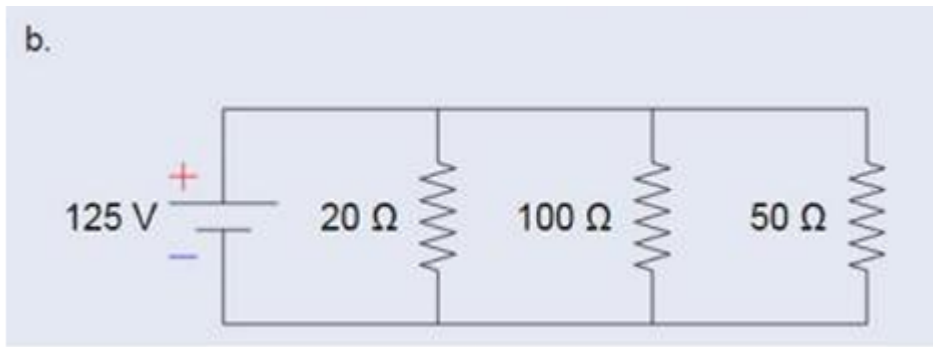
5. Power dissipated through each resistor:

$$P = VI$$

$$P_1 = 25 \times 1.25 \\ = 31.25 \text{ W}$$

$$P_3 = 62.5 \times 1.25 \\ = 78.125 \text{ W}$$

$$P_2 = 37.5 \times 1.25 \\ = 46.875 \text{ W}$$



1. Total resistance in ohms

Total resistance in parallel,  $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$\frac{1}{R_T} = \frac{1}{20} + \frac{1}{100} + \frac{1}{50}$$

$$= \frac{2}{25}$$

$$R_T = \frac{25}{2} = 12.5 \Omega$$

Given:

$$V = 125 \text{ V}$$

$$R_1 = 20 \Omega$$

$$R_2 = 100 \Omega$$

$$R_3 = 50 \Omega$$

2. Total current:

$$V = RI$$

$$I = 10 \text{ A}$$

$$125 = 12.5 I$$

$$I = \frac{125}{12.5}$$

3. Current through each resistor

$$V = RI$$

$$I_1 = \frac{125}{20} = 6.25 \text{ A}$$

$$I_2 = \frac{125}{100} = 1.25 \text{ A}$$

$$I_3 = \frac{125}{50} = 2.5 \text{ A}$$

4. Voltage drop per resistor

$$V_1 = V_2 = V_3 = V_T$$

$$= 125 \text{ V}$$

5. Power dissipated per resistor

$$P = VI$$

$$P_1 = 125 \times 6.25$$

$$= 781.25 \text{ W}$$

$$P_3 = 125 \times 2.5$$

$$= 312.5 \text{ W}$$

$$P_2 = 125 \times 1.25$$

$$= 156.25 \text{ W}$$